

QuRA: Reinforcement Learning Based Routing for Quantum Networks

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Abstract—

Quantum routing deals with identifying a set of quantum repeaters to use to create entanglement between distant end-points. Previous approaches proposed shortest-path and linear programming methods to find a solution to this problem. While the shortest path approach results in suboptimal performance, linear programming takes too long to find a solution as the network size and constraints increase. In this paper, we apply Deep Q-Reinforcement Learning (DQRL) to optimize routing in quantum networks both in terms of execution time and performance. The proposed Quantum Routing Algorithm (QuRA) first chooses which request to schedule among all requests. It then determines which route to take for the selected request. Since the number of all possible routes can be very high, we developed a hop-by-hop decision making model to lower the complexity while still attaining high performance. Experiments show that the proposed QuRA outperforms existing solutions by up to 90% in success rate and up to 79% in execution time. These results highlight QuRA as a scalable and effective solution for intelligent routing in quantum networks.

Index Terms—Quantum Network, Entanglement routing, End-to-end entanglement, Deep reinforcement learning, Fidelity-aware routing

I. INTRODUCTION

The quantum internet, promising to revolutionize secure communication [1], [2], distributed quantum computing [3], and quantum-enhanced sensing [4], relies on quantum routing to establish high-quality entanglement between distant nodes. Unlike classical routing, this task faces unique challenges: fidelity degradation from entanglement swapping (Fig. 1), probabilistic link availability, and single-use entangled links. These constraints necessitate scalable, fidelity-aware solutions for the dynamic, sequential decision problem.

Recent advances in Quantum Key Distribution (QKD) [1] and quantum processors [5] highlight the need for intelligent control. As networks grow, static or manual routing becomes impractical, increasing demand for adaptive, data-driven methods. Shortest-path methods (e.g., Q-CAST [6], qDijkstra [7]) ignore fidelity loss and network dynamics. Optimization-based approaches (e.g., REPS [8], FENDI [9]) use ILP but struggle with scalability and dynamic traffic.

RL-based approaches (e.g., DQRA [10], Purva et al. [11]) learn routing but often simplify the problem, neglecting fidelity degradation, lacking per-request decisions, or requiring retraining for topology changes. Consequently, existing models often (i) scale poorly, (ii) ignore critical quantum constraints, or (iii) rely on rigid policies ill-suited for dynamic environments

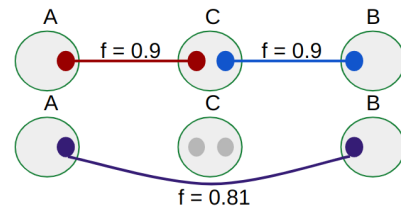


Fig. 1. Illustration of entanglement swapping, where two independent entangled links are combined to create a long-distance entangled pair with decreased fidelity (Calculation follows [12])

To address these limitations, we propose a scalable, fidelity-aware Deep Q-Reinforcement Learning (DQRL)-based Quantum Routing Algorithm (QuRA) for routing in resource-constrained quantum networks. Unlike prior methods relying on global or one-shot routing, our model makes flexible, per-hop decisions for each request based on current network state and link fidelity. This focus on fidelity-aware, real-time, per-hop decision-making under strict quantum constraints is a key distinction from existing RL solutions. This allows the agent to avoid congestion, optimize resource usage, and adapt in real time. We incorporate key quantum constraints, including fidelity decay from entanglement swapping and threshold enforcement, ensuring physically feasible, high-quality paths. Crucially, our model is trained in a topology-agnostic manner and generalizes to new networks and traffic patterns without retraining.

We evaluate our approach against state-of-the-art baselines and find that, while achieving similar success rates in fidelity-agnostic settings, our method significantly outperforms them when fidelity is considered, achieving up to 90% higher success rates and reducing execution time by up to 79%. These results show that QuRA delivers high performance with much lower computational overhead, even as network size and traffic scale, making it a practical solution for intelligent quantum routing.

Our key contributions are:

- We develop a DQRL-based Quantum Routing Algorithm (QuRA) that incrementally builds paths through fine-grained, interleaved actions across multiple requests.
- We introduce a fidelity-aware decision model that accounts for fidelity degradation due to entanglement swapping.
- We demonstrate that our model generalizes across dif-

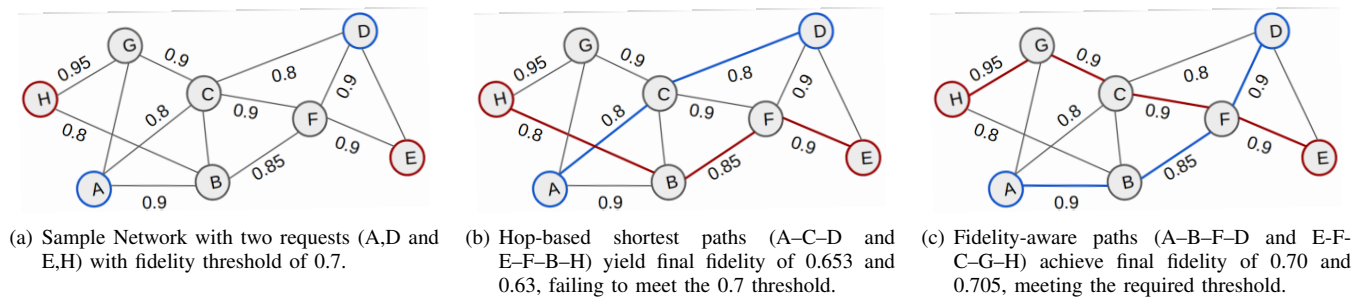


Fig. 2. Motivation example showing the limitations of hop-based routing in quantum networks. Hop based shortest paths (Fig. 2(b)) fail to meet fidelity thresholds, while longer, fidelity-aware paths (Fig. 2(c)) successfully satisfy the required fidelity for both requests.

ferent sized topologies and adapts to dynamic traffic, outperforming prior methods in scalability, flexibility, and runtime.

We use a simplified system model with abstracted link parameters to ensure tractable training and generalizability. This is a necessary foundation for future work incorporating more complex, hardware-specific constraints.

II. RELATED WORK

Multiple studies have explored routing and scheduling in quantum networks [8], [10], [13]–[15]. This section reviews major approaches and their limitations, which motivate our design. For instance, [16] introduced a performance evaluation framework based on entanglement success but did not consider concurrent routing or fidelity-aware scheduling.

To support long-distance communication, [6] proposed a four-step entanglement strategy across multiple source-destination pairs. Dijkstra-based algorithms such as qDijkstra [7] and its variants [10] use qubit availability as link weights but struggle to adapt to dynamic conditions or fidelity constraints, and scale poorly with concurrent requests. In contrast, our method learns adaptive routing policies that consider long-term effects and changing network states.

Fidelity, a key metric for entanglement quality, is addressed in works like [17]–[19]. Some use purification [20] to boost fidelity, though at the cost of additional resources, making it impractical for near-term devices. Others optimize directly based on available link fidelity. We follow this approach to maximize success rates while respecting fidelity thresholds. Still, existing works have drawbacks: [18] supports only single-request routing, [19] uses a non-scalable ILP, and [17] applies a heuristic lacking adaptability.

Fig. 2(a) shows a network with two requests, each needing minimum fidelity of 0.7. Using hop-count-based paths (Fig. 2(b)), requests (A,D) and (E,H) yield fidelities of 0.653 and 0.63, computed according to [12], respectively. In contrast, our approach (Fig. 2(c)) selects paths meeting the threshold, with fidelities of 0.70 and 0.705.

REPS [8] applies ILP to maximize throughput but becomes impractical for large networks. For fair comparison, we adapt this ILP as a baseline, though our main contribution is a scalable learning-based solution.

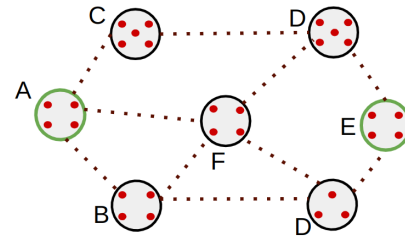


Fig. 3. A Quantum network with quantum nodes and a request. Nodes are connected by quantum channel (dotted lines), and red dots represent quantum memories at each node. Green-highlighted nodes indicate source and destination nodes for the request.

Reinforcement learning (RL) has recently been applied to routing and resource allocation. [14] uses RL to select fidelity-aware paths but only supports single-request, static scenarios. DQRA [10] focuses on scheduling via qubit availability but overlooks fidelity degradation and link constraints. [15] introduces proactive swapping via RL but focuses on local link-level optimization without global routing or scheduling across requests.

These RL models generally suffer from limited scope, handling only local or single-request decisions, and often ignore practical constraints such as entanglement generation probabilities, swapping, or fidelity thresholds. Moreover, many require retraining for new topologies or traffic patterns, limiting scalability.

To address these challenges, we introduce a scalable DQRL based Quantum Routing Algorithm (QuRA) that makes intelligent next-hop decisions for each request, dynamically avoiding congestion, and enabling smarter path selection. By enforcing key quantum constraints - fidelity decay, one-time link usage, and fidelity thresholds, our model efficiently handles multiple concurrent requests and generalizes across network topologies without retraining.

III. SYSTEM MODEL AND FRAMEWORK DESIGN

This section outlines the system model and DQRL-based framework for quantum network routing, as illustrated in Fig. 3. A quantum network uses quantum nodes as repeaters, enabling entanglement storage, transmission, and swapping. Each node contains **quantum memory** (shown as red dots) to store entangled qubits. **Quantum channels** (dotted lines in Fig. 3) connect nodes and are used to generate entangled links

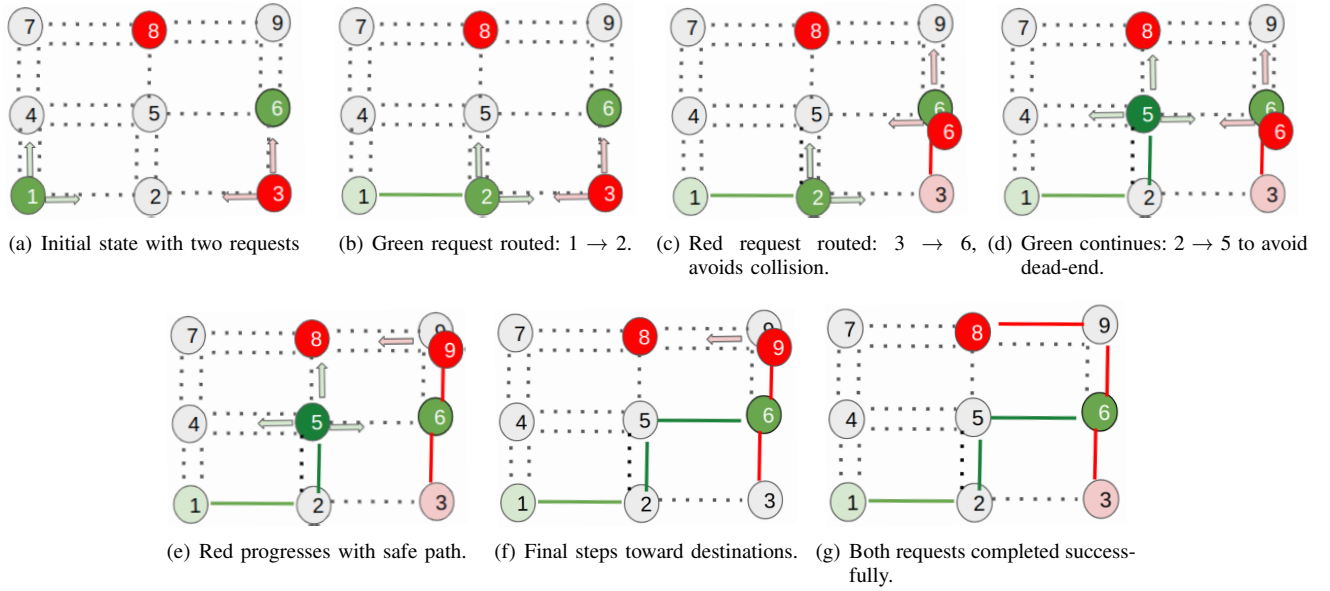


Fig. 4. Step-by-step execution of QuRA using DQRL on a 3×3 grid network. Two requests, $(1 \rightarrow 6)$ in green and $(3 \rightarrow 8)$ in red, are routed. The agent selects requests and next hops while avoiding collisions and dead ends. Solid lines represent used entangled links, and both requests are successfully completed by the final step.

(shared entangled pairs). **Fidelity** (0 to 1) quantifies the quality of an entangled state and degrades with distance and swapping. A **request** is a demand for communication (e.g., node A to E) that requires a high-fidelity entangled path (a sequence of links). All nodes act as repeaters, performing entanglement swapping to extend range.

Routing occurs over discrete **time slots** to process batches of requests. A centralized **controller** monitors the network, makes routing and scheduling decisions, and updates the state. The controller embeds a Deep Q-Reinforcement Learning (DQRL) agent that observes the global state and selects a request and its next-hop node, receiving a reward for the outcome. This step-by-step decision-making allows the agent to incrementally construct entangled paths, adapt to changing conditions, and maximize throughput without full path planning in advance.

IV. QUANTUM ROUTING ALGORITHM (QURA)

Quantum network routing involves two main tasks: (1) selecting optimal links for entanglement generation, and (2) constructing end-to-end paths for requests using the generated entanglements. Serving multiple concurrent requests under fidelity constraints makes this process challenging. In this work, we focus on the second task, scheduling and routing after successful entanglement generation. For entanglement generation, we adopt an existing strategy [21] that uses learning to select links and maximize success rates.

We introduce our deep Q-reinforcement learning based Quantum Routing Algorithm, QuRA, and demonstrate its generalization across network settings.

In QuRA the DQRL agent selects actions based on the current environment state, identifying the request and next-hop node. If no entangled link exists or the fidelity of the path,

computed as $F_{\text{swap}} = F_1 F_2 + \frac{(1-F_1)(1-F_2)}{3}$ [12], falls below a threshold or creates a loop, the request is marked as failed. Valid actions update the path and mark the link as used. Upon reaching the destination, entanglement swapping is performed. The agent then updates its policy using observed rewards and transitions, and removes completed or failed requests from the active list.

Fig. 4 illustrates QuRA on a 3×3 grid network, routing two requests: $(1 \rightarrow 6)$ (green) and $(3 \rightarrow 8)$ (red). Dashed lines represent available entangled links, while solid lines mark used ones. The agent first routes the green request from node 1 to 2. Then the red request moves from node 3 to 6, avoiding potential collisions. Next, the green request continues from node 2 to 5 to prevent getting stuck. The agent evaluates each state, selects the best request and next-hop action, and validates link availability and fidelity. By the final step (Fig. 4(g)), both requests are successfully routed.

A. Components of Deep Q-Reinforcement Learning

In QuRA, a Deep Q-Reinforcement Learning (DQRL) agent acts as the core decision-maker. Operating via a centralized controller, the agent observes the network state and selects actions that guide request routing. Through reward-based feedback, the model learns to balance fidelity, resource constraints, and dynamic conditions over time.

1) *State Representation*: At step m in time slot t , the input state $S_m^{(t)}$ includes:

- *Remaining Requests* $R_m^{(t)}$: An $M \times N$ matrix, where each row is a request vector $r = \{v_i\}$ of length N indicating the current ($v_i = 1$) and destination ($v_i = 10$) nodes.
- *Topology* $A_m^{(t)}$: Adjacency matrix $A_m^{(t)}$ representing available entangled links between neighboring nodes.

- *Edge Length* $D^{(t)}$: A $N \times N$ matrix capturing physical distances between nodes.
- *Swap Probability* $Q^{(t)}$: A vector of success probabilities for entanglement swapping at each node.

These four components form the input: $S_m^{(t)} = [R_m^{(t)}; A_m^{(t)}; D^{(t)}; Q^{(t)}]$.

2) *Action Selection and Masking*: At each step, the agent selects a (request, next-hop) pair from $M \times N$ possibilities. A binary mask $mx \in \{0, 1\}^{M \times N}$ filters invalid actions.

Each valid request $r \in \mathcal{R}_{\text{valid}}$ has a current node c_r and visited nodes $\mathcal{P}_r$. Valid next hops are:

$$\mathcal{N}_r = \{j \mid A_{c_r j} \geq 1, j \notin \mathcal{P}_r, j \neq c_r\}$$

If no valid neighbors exist for any request, a fallback mask prevents training stagnation. The agent selects the highest-Q valid action:

$$a^* = \arg \max_{a \in \mathcal{A}} (Q(s, a) \cdot mx_a) \quad (1)$$

Action Decoder: The selected action $a \in \{0, \dots, R \times N - 1\}$ maps to request r and next node j for r :

$$r = \left\lfloor \frac{a}{N} \right\rfloor, \quad j = a \bmod N \quad (2)$$

3) *Reward Function*: To encourage high-fidelity routing, the reward has both immediate and delayed components:

a) *Immediate Reward*::

$$r_m^t = \begin{cases} +10, & \text{if request completes} \\ -1, & \text{if in progress} \\ -10, & \text{if failed} \end{cases}$$

b) *Delayed Reward*:: Applied after all requests are processed:

$$R_m^t = r_m^t \cdot \lambda + N_{\text{success}} \cdot \mu + \bar{F} \cdot \nu \quad (3)$$

where:

- N_{success} : number of successful requests,
- $\bar{F}$: average fidelity of successful paths,
- λ, μ, ν : tunable hyperparameters.

This hybrid design balances short-term progress with long-term throughput and fidelity.

4) *Training the Deep Q-Network*: The agent is trained using Double DQN with experience replay and a target network to improve stability and reduce overestimation bias. During training, if the replay buffer exceeds a minimum threshold, a mini-batch of transitions is sampled to update the Q-network. The target network is periodically synchronized with the main network. To ensure valid decisions, we apply action masking before computing Q-values, preventing the agent from selecting unavailable or invalid actions.

B. Generalization Capability

To ensure scalability, we train the DQRL model using the maximum expected number of nodes and requests, fixing the input-output dimensions. During inference, the same model handles smaller networks by masking unused nodes and requests, simulating link failures or completed tasks. This enables generalization across topologies and traffic patterns without retraining or architectural changes.

V. EVALUATIONS

We evaluated our proposed algorithm (QuRA) against two baselines: a weighted shortest path approach and a modified version of REPS [8]. The original REPS maximizes total entanglement without considering fidelity. To align with our goal of establishing high-fidelity paths, we adapt REPS to enforce end-to-end routing and incorporate fidelity into its decisions.

In the weighted shortest-path approach, the weight of each edge (u, v) is based on the availability and fidelity of the entanglement link:

$$w(u, v) = \frac{w_1}{\text{capacity}(u, v)} + w_2 \cdot (-\log(\text{Fid}(u, v))) \quad (4)$$

where $\text{capacity}(u, v)$ is the number of entangled links, and $\text{Fid}(u, v)$ is the link's fidelity.

In the modified REPS-based ILP, the original throughput-focused objective is replaced by a weighted sum of the number of served requests and the logarithmic fidelity of selected links:

$$\max \left\{ w_1 \sum_i t[i] + w_2 \sum_{(u,v)} \log(\text{Fid}[i][u][v]) \right\} \quad (5)$$

where $t[i] \in \{0, 1\}$ indicates whether request i is served. To ensure optimal performance, both path selection and link assignment phases are repeated multiple times.

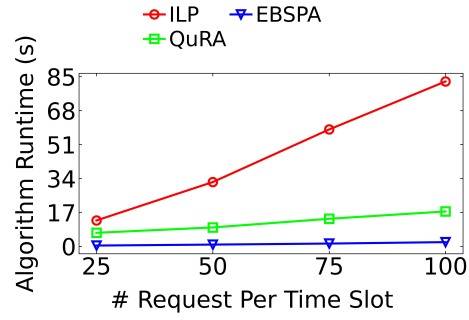
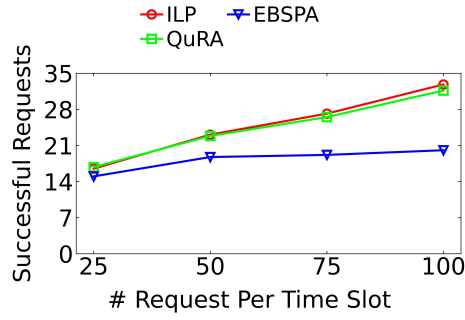
A. Evaluation Methodology

We conducted our experiments on a 10×10 grid-based quantum network spanning a 2000×2000 km area, with edge lengths varying accordingly. Each edge is initialized with 3 to 5 entangled links, and each node has 8 to 12 qubits. Entanglement swapping success probabilities are randomly drawn from $[0.85, 0.95]$. We evaluate QuRA under fidelity thresholds from 0.5 to 0.7, consistent with prior work where fidelities below 0.5 are unreliable [17].

We compared QuRA with two baselines: a fidelity-aware ILP (Fid-ILP) and an entanglement-based shortest path algorithm (EBSPA). In the first phase, we evaluated all three without fidelity constraints to assess routing success and time complexity. In the second phase, we introduced fidelity thresholds and compared the number of requests successfully served while meeting the required fidelity.

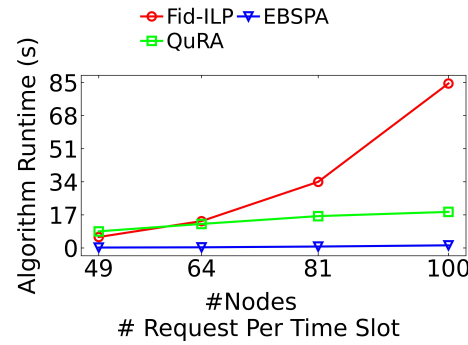
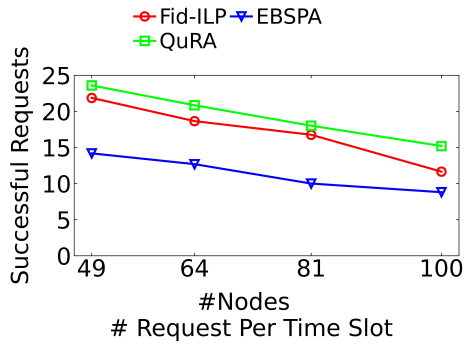
B. Evaluation Results

1) *Performance without Fidelity Constraint*: When fidelity is not considered, the ILP baseline achieves the highest success rate, closely followed by our solution QuRA. This is expected as ILP explores all possible combinations and yields an optimal solution, provided that it can find a feasible path. However, QuRA, by learning to select paths based on previous experiences, achieves performance comparable to that of ILP. Both ILP and QuRA outperform EBSPA by up to 60%, as shown in Fig. 5(a).



(a) Success rate comparison among ILP, QuRA, and EBSPA without fidelity constraints. (b) Runtime comparison showing scalability of QuRA versus ILP and EBSPA.

Fig. 5. Performance and runtime comparison of quantum routing algorithms. QuRA matches ILP's success rate while reducing runtime by 79%, outperforming EBSPA in both metrics and offering a practical solution for scalable quantum networks.



(a) Number of successful requests decreases with increasing network size, but QuRA consistently outperforms Fid-ILP and EBSPA. (b) Runtime comparison shows Fid-ILP's exponential growth, while QuRA scales efficiently with approximately 80% lower computation time.

Fig. 6. Performance evaluation of QuRA, Fid-ILP, and EBSPA across varying network sizes with the number of requests equal to the number of nodes and a fidelity threshold of 0.7.

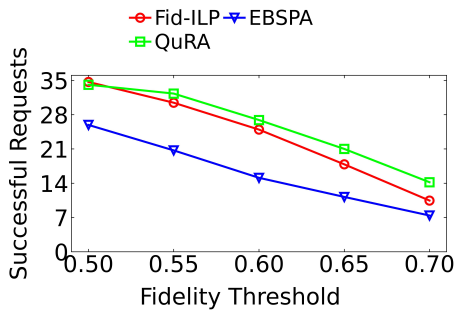


Fig. 7. Success rate comparison under varying fidelity thresholds. QuRA outperforms Fid-ILP and EBSPA, with increasing advantage at higher thresholds by effectively learning to avoid low-fidelity links.

Despite its high performance, ILP suffers from significant run-time overhead. As illustrated in Fig. 5(b), the runtime of ILP increases rapidly with the number of requests, making it unsuitable for large-scale or real-time applications. In contrast, EBSPA exhibits nearly constant runtime, while QuRA offers a balanced trade-off, achieving performance close to ILP but with significantly lower computation time. Specifically, QuRA reduces runtime by approximately 79.36% compared to ILP.

2) *Performance with Fidelity Constraint:* We also compare the methods under increasing fidelity constraints. As depicted in Fig. 7, for low fidelity thresholds, the performance trends remain similar: ILP and QuRA outperform EBSPA, and the gap remains consistent. However, as the fidelity threshold increases, QuRA consistently outperforms both ILP and EBSPA, by up to 40% and 90%, respectively. This is because QuRA can adapt its policy to meet strict fidelity constraints by learning to avoid low-fidelity links, whereas ILP and EBSPA struggle to accommodate the additional constraints.

The runtime behavior with fidelity constraints remains consistent with the previous observation. While ILP becomes increasingly computationally expensive with complex constraints, QuRA maintains its efficiency due to its learned policy structure.

3) *Performance for different network size:* In Fig. 6, we compare the performance of our QuRA against ILP and EBSPA across different network sizes and varying numbers of requests per time slot. For consistency, the number of requests is set equal to the number of nodes in the network. A fidelity threshold of 0.7 is used for this scenario. As shown in Fig. 6(a), the number of successful requests decreases as the network size increases. This is because longer paths

are required in larger networks, leading to greater fidelity loss due to entanglement swapping. Despite this, our method consistently outperforms both ILP and EBSPA in terms of successful request completion.

Fig. 6(b) illustrates the runtime performance. While ILP performs better in small networks, its runtime grows exponentially with increasing network size and number of requests. In contrast, our method exhibits linear growth in runtime, achieving approximately 80% reduction in computation time compared to ILP for larger networks. These results highlight the scalability and efficiency of our approach, particularly in large-scale quantum networks.

C. Training Overhead

The DQRL model in QuRA, with approximately 2.5 million parameters, is trained from scratch over 60 hours on an NVIDIA T4 GPU. It handles 100 nodes and 100 requests per episode, with around 1000 steps per episode. The model uses two hidden layers, the Adam optimizer (learning rate 0.0001), a batch size of 2048, and a replay buffer of 100,000 to ensure stable learning over a large state-action space. Despite the training time, the model generalizes well across varying network conditions, such as link failures and fluctuating request loads, without retraining.

VI. CONCLUSION

We presented a scalable, fidelity-aware DQRL based Quantum Routing Algorithm (QuRA) for efficient routing and scheduling in quantum networks. QuRA enables intelligent, step-by-step decision making to jointly serve multiple requests while dynamically selecting collision-free paths within a single episode. Using DQRL, our approach achieves higher request success and reduced runtime compared to traditional routing methods. The significance of our work lies in its ability to make fine-grained decisions under fidelity constraints, improving scalability and adaptability.

Although our current evaluation utilizes simplified grid topologies and abstract parameters to focus on the core routing challenge, and the model assumes a centralized controller with global state access, QuRA offers a solid, fidelity-aware foundation for coordination. In future work, we aim to extend QuRA to distributed inference for decentralized, collision-free coordination, and incorporate more realistic noisy channels and hardware constraints to strengthen its practical applicability in near-term quantum networks.

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REFERENCES

- [1] P. Jouguet, S. Kunz-Jacques, A. Leverrier, P. Grangier, and E. Diamanti, "Experimental demonstration of long-distance continuous-variable quantum key distribution," *Nature photonics*, vol. 7, no. 5, pp. 378–381, 2013.
- [2] T. Islam and E. Arslan, "Quantum key distribution with single qubit transmission," in *2024 International Conference on Quantum Communications, Networking, and Computing (QCNC)*. IEEE, 2024, pp. 357–358.
- [3] H. Buhrman and H. Röhrig, "Distributed quantum computing," in *International Symposium on Mathematical Foundations of Computer Science*. Springer, 2003, pp. 1–20.
- [4] T. Chalopin, C. Bouazza, A. Evrard, V. Makhlov, D. Dreon, J. Dalibard, L. A. Sidorenkov, and S. Nascimbene, "Quantum-enhanced sensing using non-classical spin states of a highly magnetic atom," *Nature communications*, vol. 9, no. 1, p. 4955, 2018.
- [5] D. Gao, D. Fan, C. Zha, J. Bei, G. Cai, J. Cai, S. Cao, F. Chen, J. Chen, K. Chen *et al.*, "Establishing a new benchmark in quantum computational advantage with 105-qubit zuochongzhi 3.0 processor," *Physical Review Letters*, vol. 134, no. 9, p. 090601, 2025.
- [6] S. Shi and C. Qian, "Concurrent entanglement routing for quantum networks: Model and designs," in *Proceedings of the Annual conference of the ACM Special Interest Group on Data Communication on the applications, technologies, architectures, and protocols for computer communication*, 2020, pp. 62–75.
- [7] R. Van Meter, T. Satoh, T. D. Ladd, W. J. Munro, and K. Nemoto, "Path selection for quantum repeater networks," *Networking Science*, vol. 3, pp. 82–95, 2013.
- [8] Y. Zhao and C. Qiao, "Redundant entanglement provisioning and selection for throughput maximization in quantum networks," in *IEEE INFOCOM 2021-IEEE Conference on Computer Communications*. IEEE, 2021, pp. 1–10.
- [9] H. Gu, Z. Li, R. Yu, X. Wang, F. Zhou, J. Liu, and G. Xue, "Fendi: Toward high-fidelity entanglement distribution in the quantum internet," *IEEE/ACM Transactions on Networking*, 2024.
- [10] L. Le and T. N. Nguyen, "Dqra: Deep quantum routing agent for entanglement routing in quantum networks," *IEEE Transactions on Quantum Engineering*, vol. 3, pp. 1–12, 2022.
- [11] P. Sharma, S. Gupta, V. Bhatia, and S. Prakash, "Deep reinforcement learning-based routing and resource assignment in quantum key distribution-secured optical networks," *IET Quantum Communication*, vol. 4, no. 3, pp. 136–145, 2023.
- [12] W. Kozłowski, A. Dahlberg, and S. Wehner, "Designing a quantum network protocol," in *Proceedings of the 16th international conference on emerging networking experiments and technologies*, 2020, pp. 1–16.
- [13] C. Ciconetti, M. Conti, and A. Passarella, "Request scheduling in quantum networks," *IEEE Transactions on Quantum Engineering*, vol. 2, pp. 2–17, 2021.
- [14] J. Roik, K. Bartkiewicz, A. Černoč, and K. Lemr, "Routing in quantum communication networks using reinforcement machine learning," *Quantum Information Processing*, vol. 23, no. 3, p. 89, 2024.
- [15] T. Islam, M. Arifuzzaman, and E. Arslan, "Reinforcement learning based proactive entanglement swapping for quantum networks," in *2024 International Conference on Quantum Communications, Networking, and Computing (QCNC)*. IEEE, 2024, pp. 135–142.
- [16] Y. Lee, W. Dai, D. Towsley, and D. Englund, "Quantum network utility: A framework for benchmarking quantum networks," *Proceedings of the National Academy of Sciences*, vol. 121, no. 17, p. e2314103121, 2024.
- [17] K. Li, V. Chaudhary, S. G. Sanchez, and K. R. Chowdhury, "Q-firm: Fidelity-based rate maximizing routes for quantum networks," in *2023 IEEE 20th Consumer Communications & Networking Conference (CCNC)*. IEEE, 2023, pp. 775–780.
- [18] M. Caleffi, "Optimal routing for quantum networks," *Ieee Access*, vol. 5, pp. 22 299–22 312, 2017.
- [19] D. H. Nguyen, E. Hunt, D. J. Horton, T. N. Nguyen, and B.-H. Liu, "Maximizing entanglement routing rate in quantum networks: Approximation algorithms," *IEEE Transactions on Network Science and Engineering*, 2025.
- [20] H. Hu, H. Lun, Z. Deng, J. Tang, J. Li, Y. Cao, Y. Wang, Y. Liu, D. Wu, H. Yu *et al.*, "High-fidelity entanglement routing in quantum networks," *Results in Physics*, vol. 60, p. 107682, 2024.
- [21] T. Islam, M. Arifuzzaman, and E. Arslan, "Adaptive Entanglement Generation for Quantum Routing," *arXiv preprint arXiv:2505.08958*, 2025.